

A Two-Stage Product Recommender on H&M Data

Capstone, Pillar 5: eCommerce recommendations

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Problem: pick the 12 products each customer buys next week

Business

- 105k products, a shopper sees a few dozen
- “You may also like”, app home and emails default to the same best sellers
- Goal: personalize those 12 slots, give smaller items a chance

Task type: recommendation, solved as learning to rank

1. Candidate generation (recall): 7 recommenders, ~ 80 items/customer
2. Ranking: binary classifier $P(\text{buy}_{c,i,w})$, sort, take top 12

Primary metric: MAP@12 (Kaggle definition)

$$\text{AP@12}(c) = \frac{1}{\min(m_c, 12)} \sum_{k=1}^{12} P(k) \text{rel}(k)$$

Also: Recall@12, NDCG@12, hit rate, coverage, AUC, candidate recall

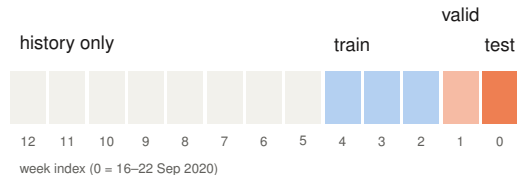
Business KPIs: CTR and add-to-cart on slots, conversion uplift vs best sellers (A/B), items per order, long-tail share of slots, hit-rate disparate impact ≥ 0.8 per age band

Data: H&M Personalized Fashion Recommendations (Kaggle 2022)

Table	Rows	Content
transactions	31.8M	day, customer, article, price, channel
customers	1.37M	age, club, newsletter flags
articles	105.5k	11 product attributes + text

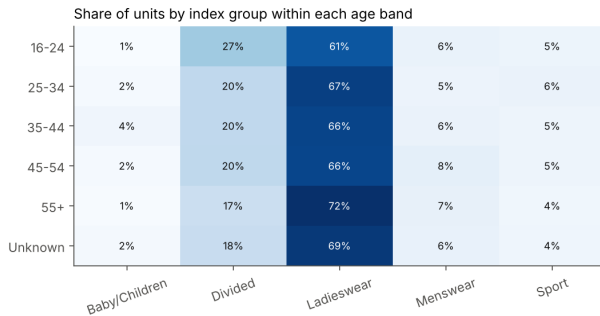
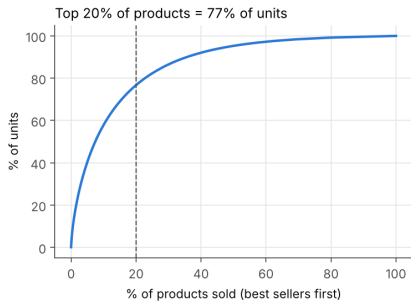
- Row counts verified against the official competition
- **Sample:** last 13 weeks, customers whose hashed id ends in 00–33 (20.3%, deterministic)
- 789k transactions, 106.7k buyers, 31.2k products
- Missing: FN/Active 65% (= no), age 1.2%, club 0.4%
- 8.4% exact duplicate rows = multi-unit lines (kept)
- Price IQR outliers 3.9% (real items, kept)

Time split (no leakage: features use only the 8 weeks before the target week)



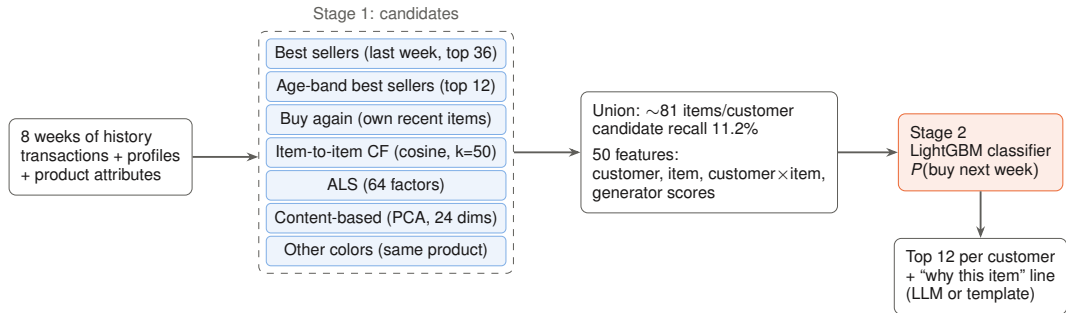
- Train labels: weeks 2–4 (1.02M candidate rows)
- Validation: week 1 (tuning, model choice)
- Test: week 0, 14,023 buyers, scored once

EDA: concentrated sales, age-dependent taste, low repeat rate



- Top 20% of products = 77% of units; store sales stop Mar–May 2020 (COVID), online is 70%
- Only **2.6%** of next-week items were bought by the same customer in the prior 8 weeks; 4.5% are another color of an owned product; 38% of buyers have no 8-week history
- Young buyers repeat less: 2.0% repeat items for 16–24 vs 2.8% for 55+ → expect a quality gap

Pipeline



Features, selection and dimensionality reduction

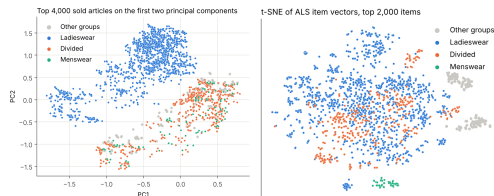
50 features, all from the 8 weeks before the target week

- Generator scores/ranks (11), item (11), customer (13), customer $\times$ item (10), item categoricals (5)
- Domain: other-color count, trend (1w vs 4w avg), price ratio, taste shares, age gap
- LR: impute + missing flags + scale + one-hot; LightGBM: native categoricals

Selection (same LightGBM, validation week)

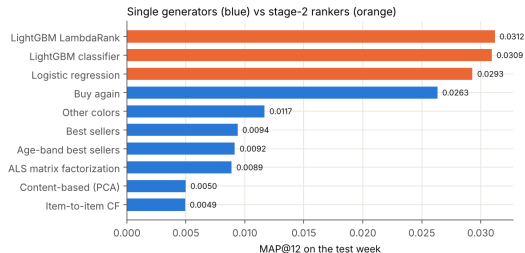
Set	Feat.	MAP@12
All features	50	0.0301
Mutual information top 25 + cat.	30	0.0303
L1 logistic (C=0.01) + cat.	27	0.0291

Kept all 50: MI is univariate and drops 12 of 13 customer features; needed age features in-model for the fairness tests



- **PCA**: 600 one-hot product columns $\rightarrow$ 24 components (60% variance), used as item vectors for content-based candidates
- **t-SNE** of ALS item vectors: clusters by index group without seeing attributes $\rightarrow$ purchases carry style signal

Results on the held-out test week (14,023 customers)



Model	MAP@12	R@12	Hit	AUC
Best sellers	0.0094	0.028	7.1%	—
Buy again	0.0263	0.056	11.2%	—
Logistic regression	0.0293	0.062	12.0%	0.730
LightGBM classifier	0.0309	0.066	13.1%	0.758
LightGBM LambdaRank	0.0312	0.065	12.7%	0.756

- Selected on validation MAP@12 (0.0305 vs 0.0301 LambdaRank)
- 3.3× best sellers; +17% over the best single generator
- LambdaRank ties MAP but loses recall, hit rate, AUC
- Context: a top-2% Kaggle team (45/3,006) scored 0.0300 on full data
- Warm customers 0.0448 vs cold 0.0086 (cold $\approx$ best sellers)

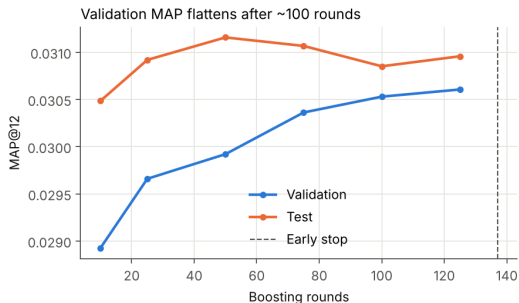
Tuning and overfitting

Tuning: random search, 8 LightGBM settings, early stopping on validation AUC (first metric only)

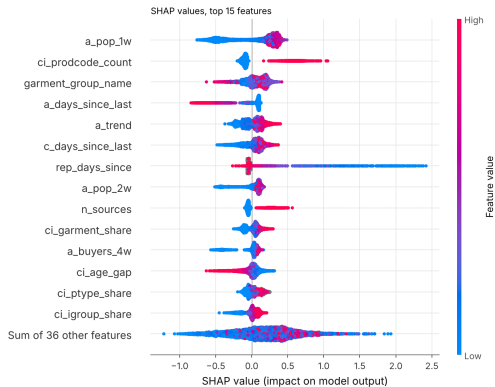
Best: 127 leaves, lr 0.05, min child 200, colsample 0.7, subsample 0.6, $\lambda_2=5$, 137 rounds

Split	MAP@12	Hit rate	AUC
Train (weeks 2–4)	0.0420	16.0%	0.856
Validation (week 1)	0.0305	12.7%	0.768
Test (week 0)	0.0309	13.1%	0.758

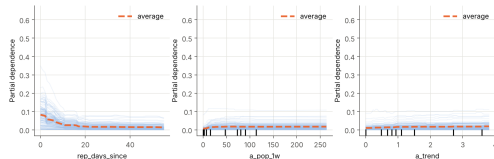
- Train–valid gap: trees memorize training weeks
- Valid $\approx$ test: generalizes to an unseen week
- Imbalance: 0.43% positives in test $\rightarrow$ ranking metrics, not accuracy



Explainability: exact TreeSHAP, PDP and ICE

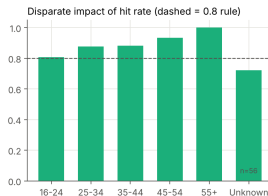
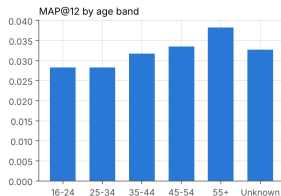


PDP (orange) and ICE (blue) on predicted purchase probability



- Top drivers: last week's sales, owning another color of the product, garment group, item and customer recency, trend
- Bought the exact item recently: rare but the largest single effect (up to +2.4 log-odds); gain ranks it 1st, SHAP 7th
- Age gap is 12th → how age enters the model
- SHAP chosen over LIME: exact and stable for trees

Bias audit: age bands (customer side) and popularity (item side)



Band	MAP@12	Hit	DI	TPR
16-24	0.0283	12.0%	0.81	0.450
25-34	0.0283	13.0%	0.88	0.434
35-44	0.0317	13.1%	0.88	0.429
45-54	0.0335	13.9%	0.93	0.475
55+	0.0382	14.8%	1.00	0.518

Customer side (only age is available; gender not inferred, postal code dropped)

- Quality rises with age; 16–24 passes the 0.8 rule only just
- Demographic parity diff. 0.012 (selection 14.1–15.3%)
- Equalized odds: TPR gap 0.089, FPR gap 0.012
- Unknown band (n=56) too small to judge

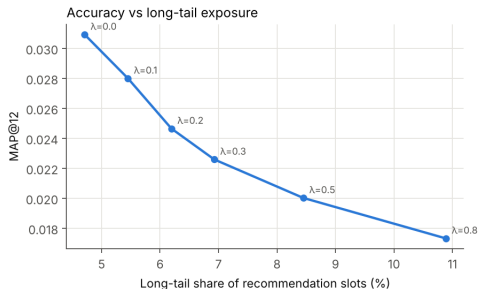
Item side

- Head (top 20% sellers) gets 95.3% of slots; long tail is 34.1% of purchases but 4.7% of slots
- Exposure Gini 0.90

Mitigations

Variant	MAP@12	Min DI	Gap
Baseline ranker	0.0309	0.81	0.0099
A: reweight age bands	0.0304	0.81	0.0118
B: drop age features	0.0307	0.86	0.0096
C: re-rank $\lambda=0.1$	0.0280	—	—

- A (pre-processing) did not help: young bands already the largest
- B (unawareness): -0.8% MAP, min DI $0.81 \rightarrow 0.86$
- C (post-processing): long tail $4.7\% \rightarrow 5.5\%$ of slots, coverage $29\% \rightarrow 32\%$, but -9% MAP
- **Ship B**; give long tail 1–2 dedicated slots instead of re-ranking; monitor DI weekly



Deployment, GenAI, limitations, next steps

Serving: nightly batch scoring → FastAPI lookup (/recommendations, /similar, /health), demo page, Docker, CI (ruff + 19 tests), MLflow, config-driven runs, monitoring + rollback plan

GenAI: ranker features → facts → Claude writes a ≤ 16 -word reason; guardrail blocks personal traits; template fallback without a key

Limitations: candidate recall ceiling 11.2%; 20% sample; 38% cold buyers; scaled prices, no margin; offline $\neq$ online

You may also like

Two-stage recommender demo on H&M data. Recommendations use data up to Sep 15, 2020.

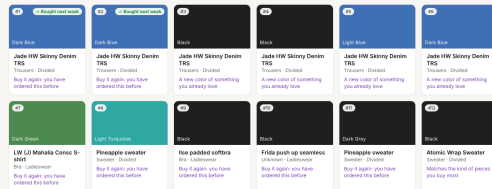
Demo customer 1 or paste a customer_id Recommend All reasons

Customer R57364873... Past items 8 Bought next week 2 of 12

Recently bought last 8 weeks



Recommended for you candidates + LightGBM ranker



Green badge = the customer really bought that item in the following week (16 to 22 Sep 2020), which the model never saw.

Next: more candidate sources (longer window, bought-together pairs, trending in category), full data, ship variant B, A/B test vs best sellers